

# Capstone Project

## Walmart Weekly Sales

Intellipaat

Executive-Post-Graduate-Certification-in-Data-Science-and-Artificial-Intelligence

Date- Oct 2024



# Contents

|          |                                                   |           |                                     |
|----------|---------------------------------------------------|-----------|-------------------------------------|
| <b>1</b> | Problem Statement                                 | <b>9</b>  | Inferences from the Same            |
| <b>2</b> | Project Objective                                 | <b>10</b> | Future Possibilities of the Project |
| <b>3</b> | Data Description                                  | <b>11</b> | Conclusion                          |
| <b>4</b> | Data Pre-processing Steps and Inspiration         |           |                                     |
| <b>5</b> | Choosing the Algorithm for the Project            |           |                                     |
| <b>6</b> | Motivation and Reasons For Choosing the Algorithm |           |                                     |
| <b>7</b> | Assumptions                                       |           |                                     |
| <b>8</b> | Model Evaluation and Techniques                   |           |                                     |

# The Problem Statement

Walmart is an American multinational retail corporation that operates a chain of hypermarkets, discount department stores, and grocery stores in the United States.

It has been facing issues in managing inventory - to match the demand with respect to supply.

It has requested Intellipaart to come up with useful insights using their data and make prediction models to forecast the sales for 12 to 16 weeks.

It has provided last 3 years weekly sales data containing 6435 rows and 8 columns to enable data prediction.



# Project Objective

## Pre- processing

- Shape of data
- Data type of columns
- Handle Null values
- Handle Duplicates values
- Analyze Unique values for each column

## Insights

- Identify store with Max and Min sales
- Identify Top 5 and Bottom 5 stores by Sales
- Diff between Max and Min Sales
- Identify Week of Max sales across all stores
- Graph- Sales by Store Number

## Features

- Heat map of all features with one another
- Identify and remove features having outliers
- Analyze impact of Annual Year, Unemployment and CPI on Sales
- Correlation of Unemployment & CPI with Sales

## Predictive models

- Analyze what Machine Learning will fit the requirement
- Analyze Time Series- ARIMA and SARIMAX
- Present sales predictions using time series graph

# Pre Processing

# Pre- processing

- **Dataset Overview:**
  - **6435** rows
  - **8** columns
- **Data Types:**
  - **Store-** Store number [**int64** ]
  - **Date-** Week of Sales [**object**]
  - **Weekly\_Sales-** Sales for the given store in that week [**float64**]
  - **Holiday\_Flag-** if it is a holiday week [**int64**]
  - **Temperature-** Temperature on the day of the sale [**float64**]
  - **Fuel\_Price-** Cost of the fuel in the region [**float64**]
  - **CPI-** Consumer Price Index [**float64**]
  - **Unemployment-** Unemployment **Rate**[**float64**]

# Pre- processing

- **Null and Duplicates:**
  - There are **No Null nor Duplicate** values in the Dataset
- **Number of Unique values:**
  - Store 45
  - Date 143
  - Weekly\_Sales 6435
  - Holiday\_Flag 2
  - Temperature 3528
  - Fuel\_Price 892
  - CPI 2145
  - Unemployment 349

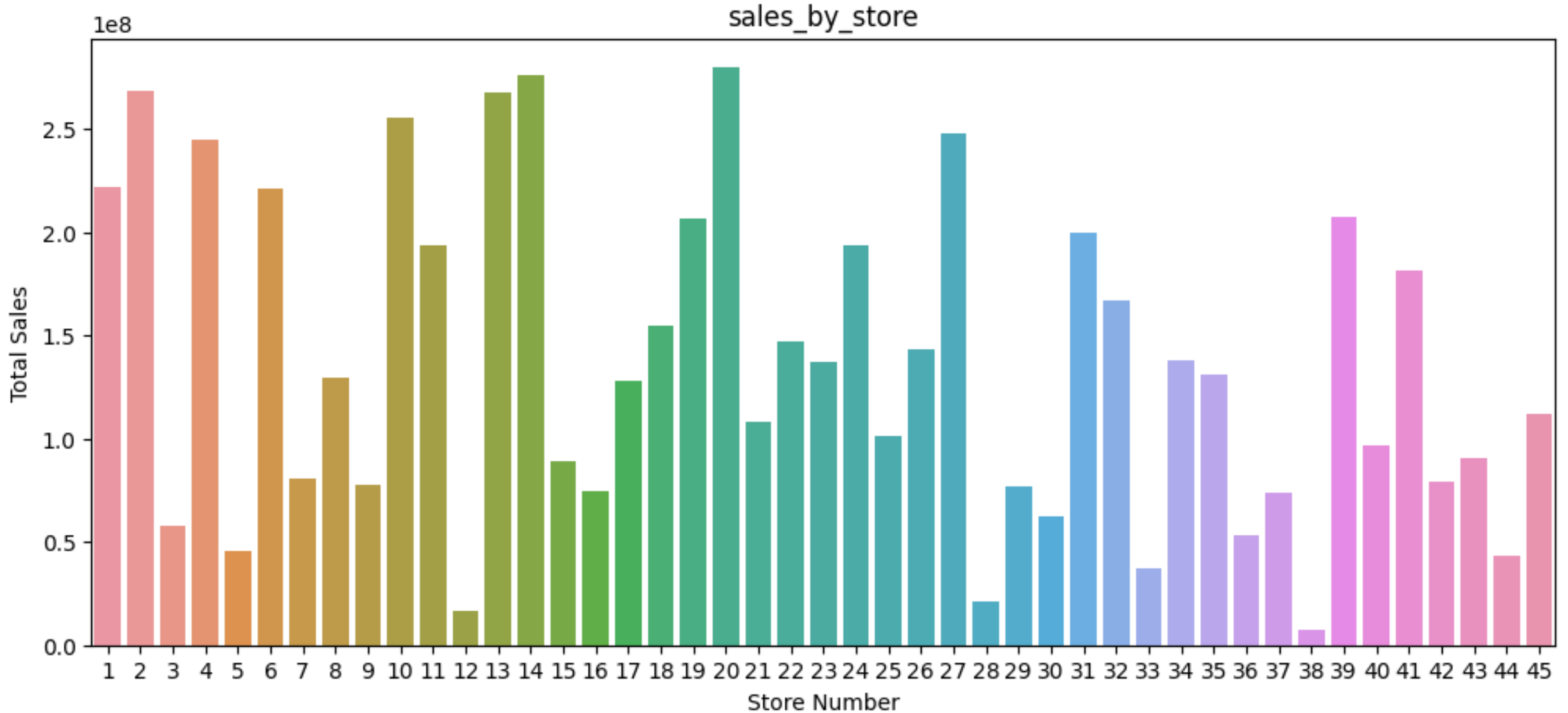
# Insights



# Insights

- Store number 14 was found to have max sales= USD 3818686.45 in the week ending 24 Dec 2010
- Top 5 stores for Sales are - 14, 20,10, 4, 13
- Store number 33 was found to have max sales= USD 209986.25 in the week ending 3 Dec 2010
- Bottom 5 stores for Sales are – 33, 33, 33, 33, 33 [Interestingly all the 5 the bottom slot was taken by Store 33]
- Difference between maximum sales and minimum sales- USD 3608700
- Weeks for max sales across all the stores combined by Total Sales were
  - **24 Dec 2020**
  - **23 Dec 2011**
  - **25 Nov 2011**
  - **26 Nov 2010**
  - **17 Dec 2010**

# Insights- Sales by Store Number



# Features

# Features- Heat Map

Insignificant +ve and -ve correlations found for external factors with Weekly Sales



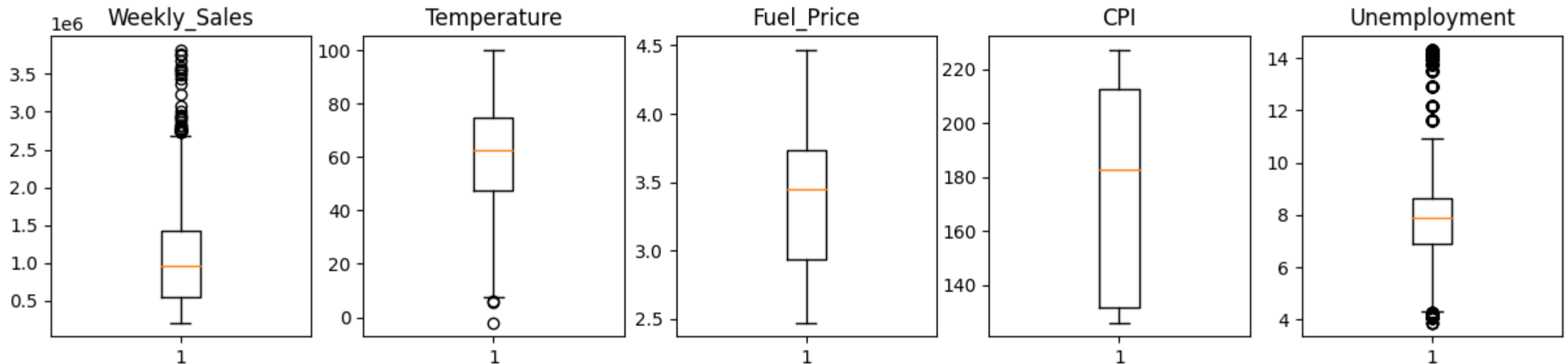
## Correlation of Weekly\_Sales with

1. Holiday Flag:
  - 0.037 (low +ve correlation)
2. Temperature:
  - -0.064 (low -ve correlation)
3. Fuel Price:
  - 0.0095 (very low +ve correlation)
4. CPI:
  - -0.073 (low -ve correlation)
5. Unemployment:
  - -0.11 (low -ve correlation)

# Features- Outliers

Outliers were found and removed from the following columns using the IQR Method

1. Weekly Sales
2. Temperature
3. Unemployment

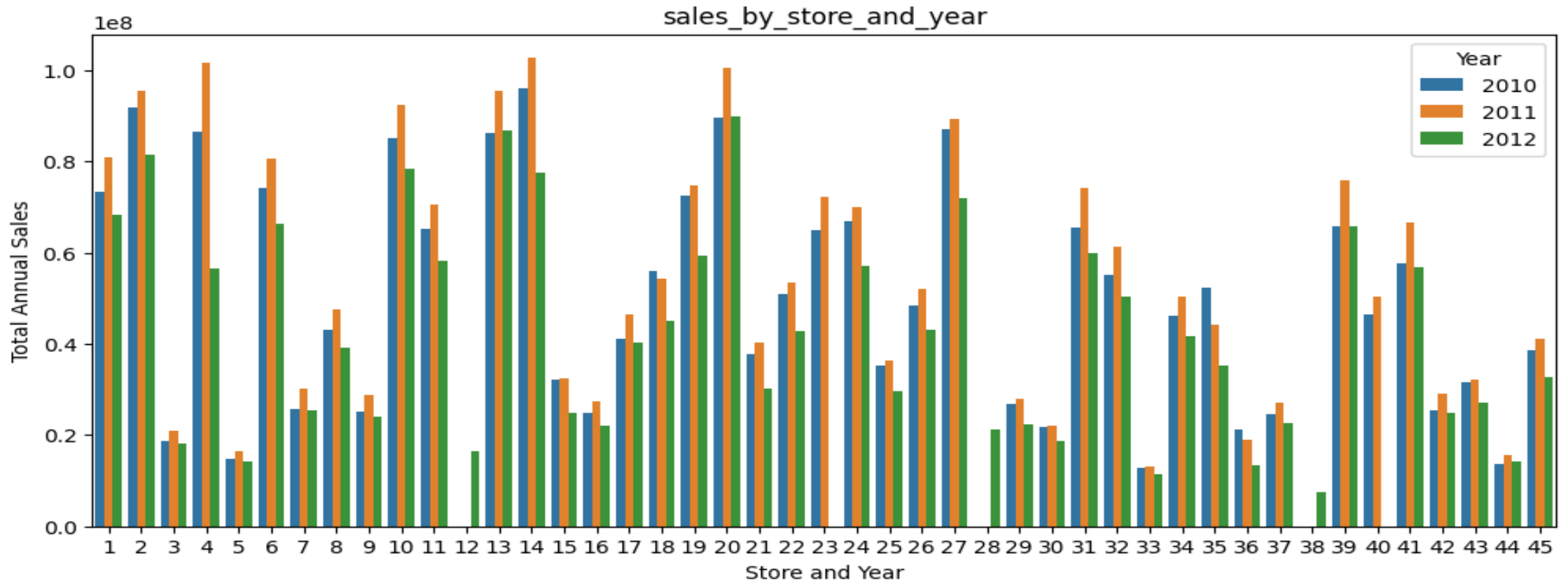


# Features- Impact of Annual Year on Sales

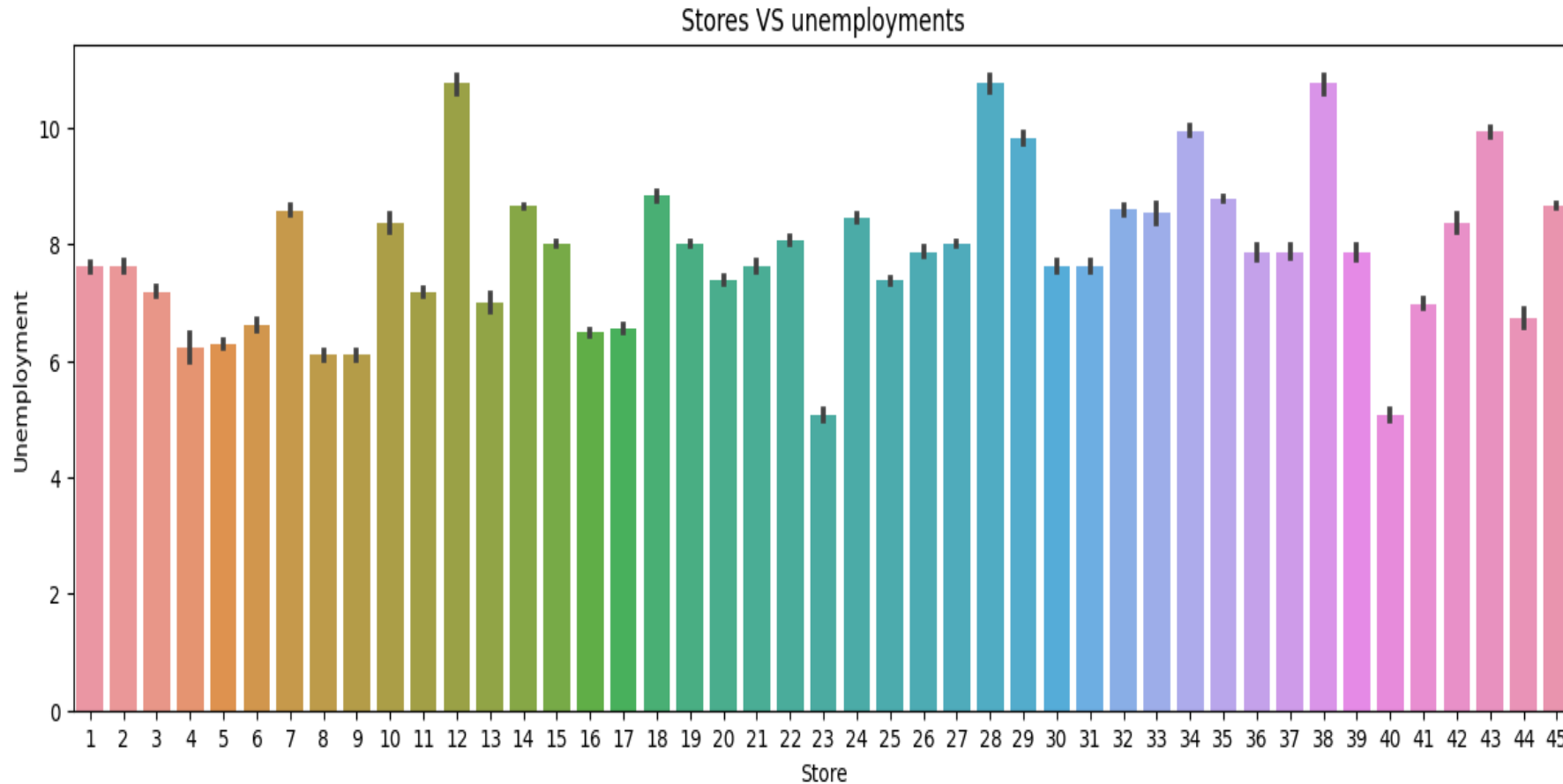
Sales were found to be high for year 2011.

This is because the sales data is available fully for 2011.

However sales data is only partially available for 2010 and 2012

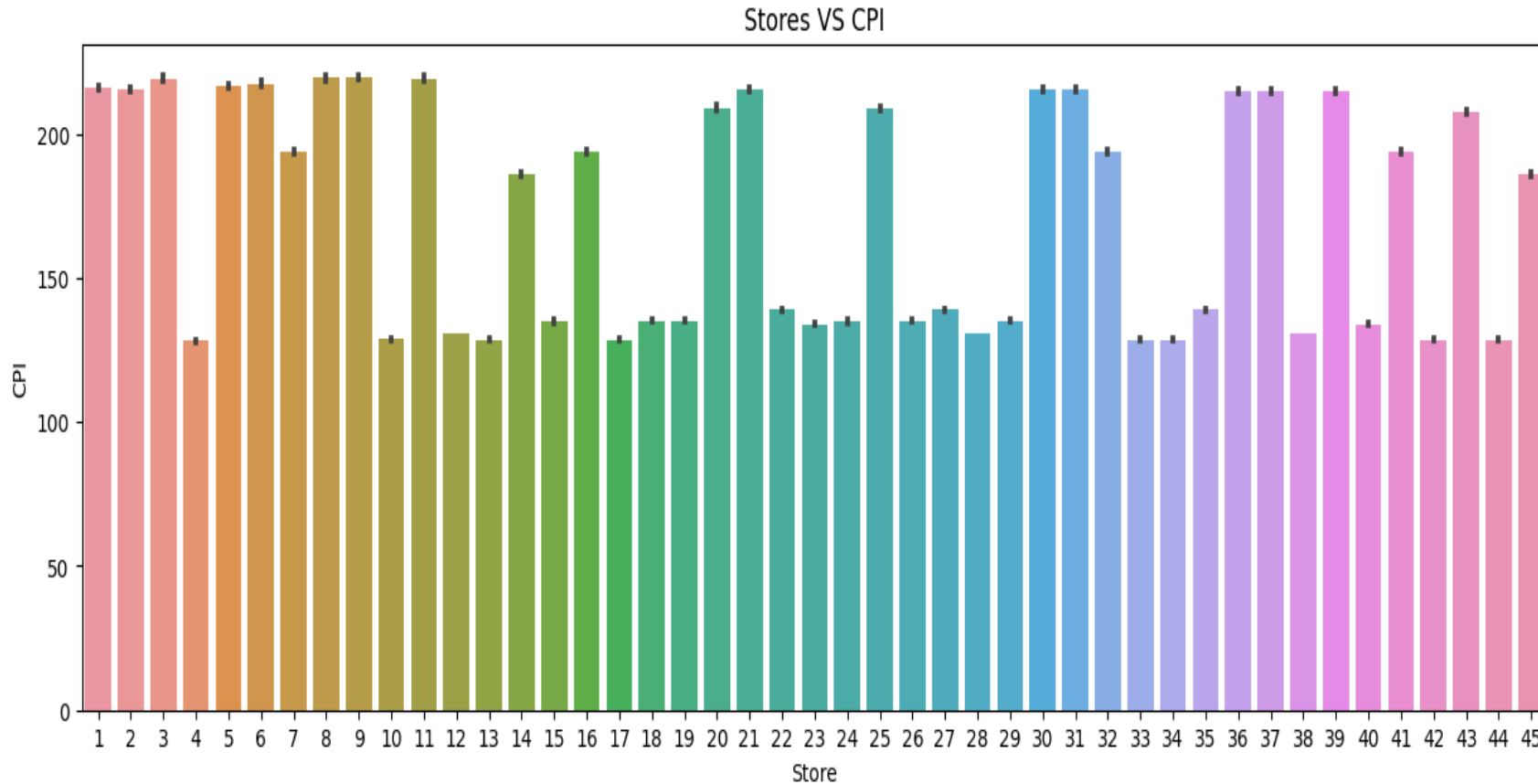


# Features- Impact of Unemployment on Sales



- Store 40 and 23 area had minimum unemployment
- Store 12, 28, 38 area had maximum unemployment
- Store 44- had maximum negative correlation with unemployment
  - So, with increase in unemployment Store 44 had max reduction in Sales
- Stores 36- had maximum positive correlation with unemployment
  - So, with increase in unemployment Store 36 had max increase in Sales

# Features- Impact of CPI on Sales



- Store 1,2,3,5,6,8,9,11, 30,31 had max increase in CPI
- Store 38 had max negative correlation
  - With increase in CPI, sales of store 38 reduced maximum
- Store 44 had max positive correlation
  - With increase in CPI, sales of store 44 also increased maximum



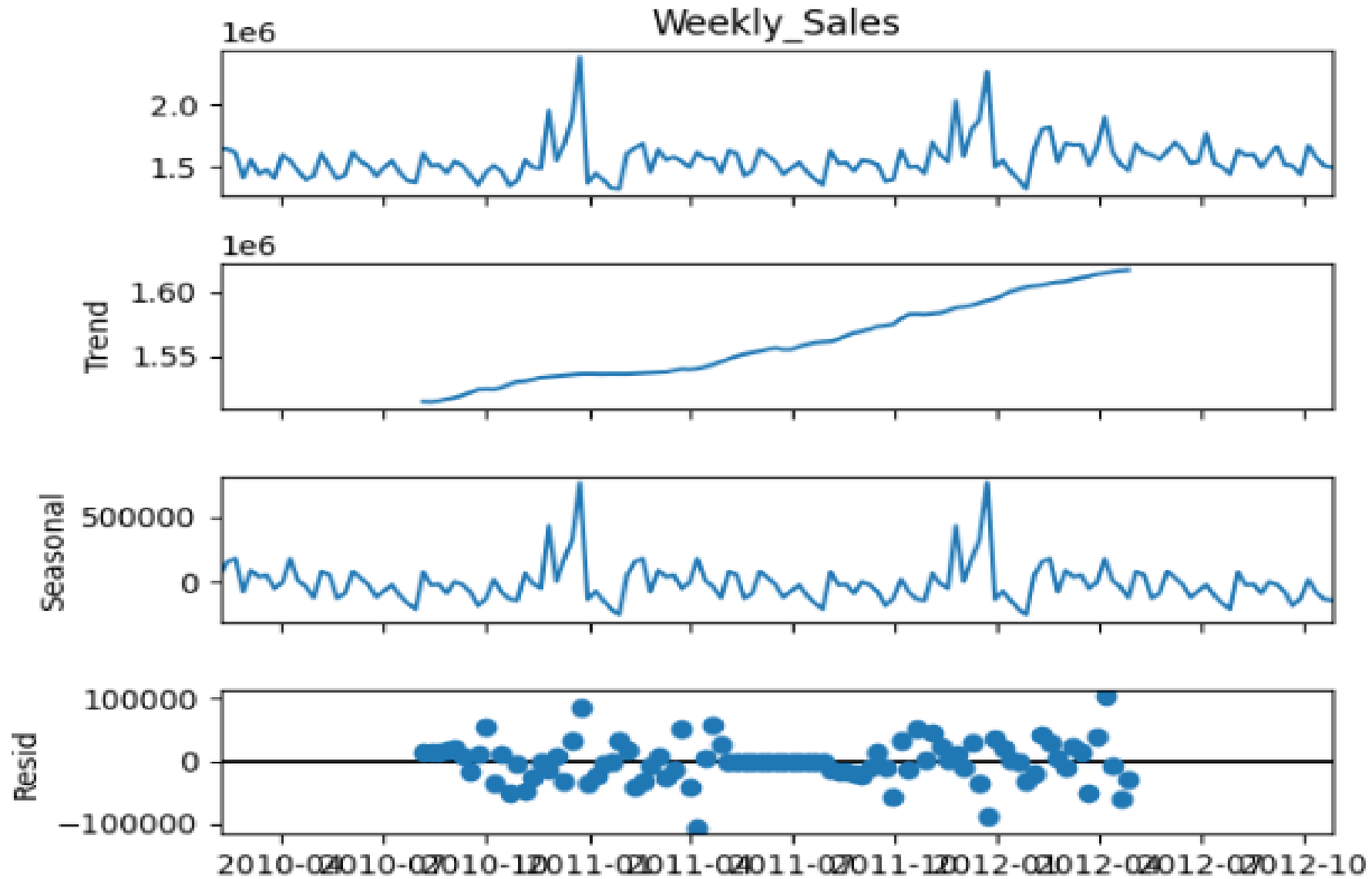
# Prediction Models

# Predictive Models

## Choosing the right model

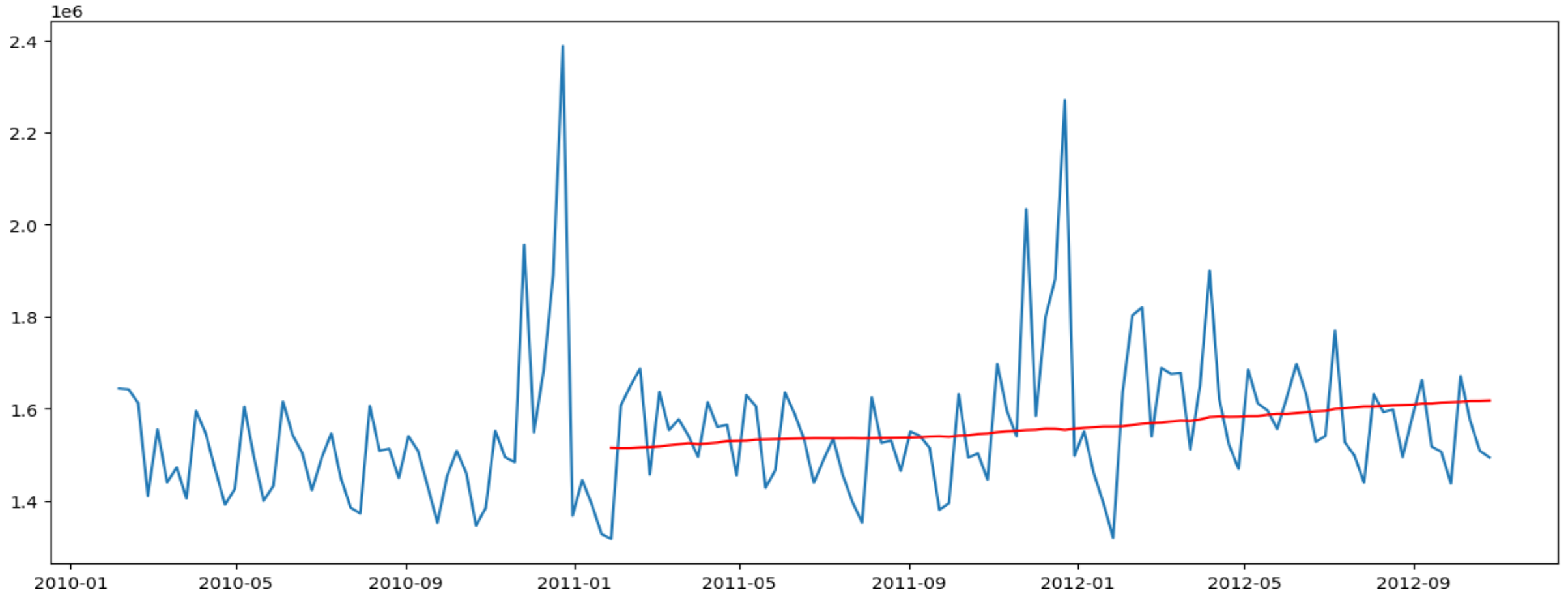
- Data set has columns like CPI, unemployment, temperature, holiday and fuel price.
- We do not have future trend of CPI, unemployment, temp, holiday and fuel price columns for 12 to 16 weeks.
- Prediction of Weekly Sales can not be done using traditional ML Algorithms such as Linear Regression, Decision Tree, Random Forest since they will need these column details completely.
- Future prediction needs to be done only basis time- hence this becomes a Time Series ML Algo
- We tested ARIMA and SARIMAX, however SARIMAX gives a better representation due to seasonality of the data.
- Following assumptions & tests were done on the data:
  - Assumed seasonality window = 52 weeks
    - since the plot of mean data gives a straight line with window =52
  - Conducted stationarity test of Weekly\_Sales
    - It was found stationary
  - Conducted ACF & PACF Tests : found q value = 2, p value= 2, d value = 0
  - ARIMA was not accurate
  - SARIMAX gave a strong representation of the data

# Looking at trend, seasonality and residue

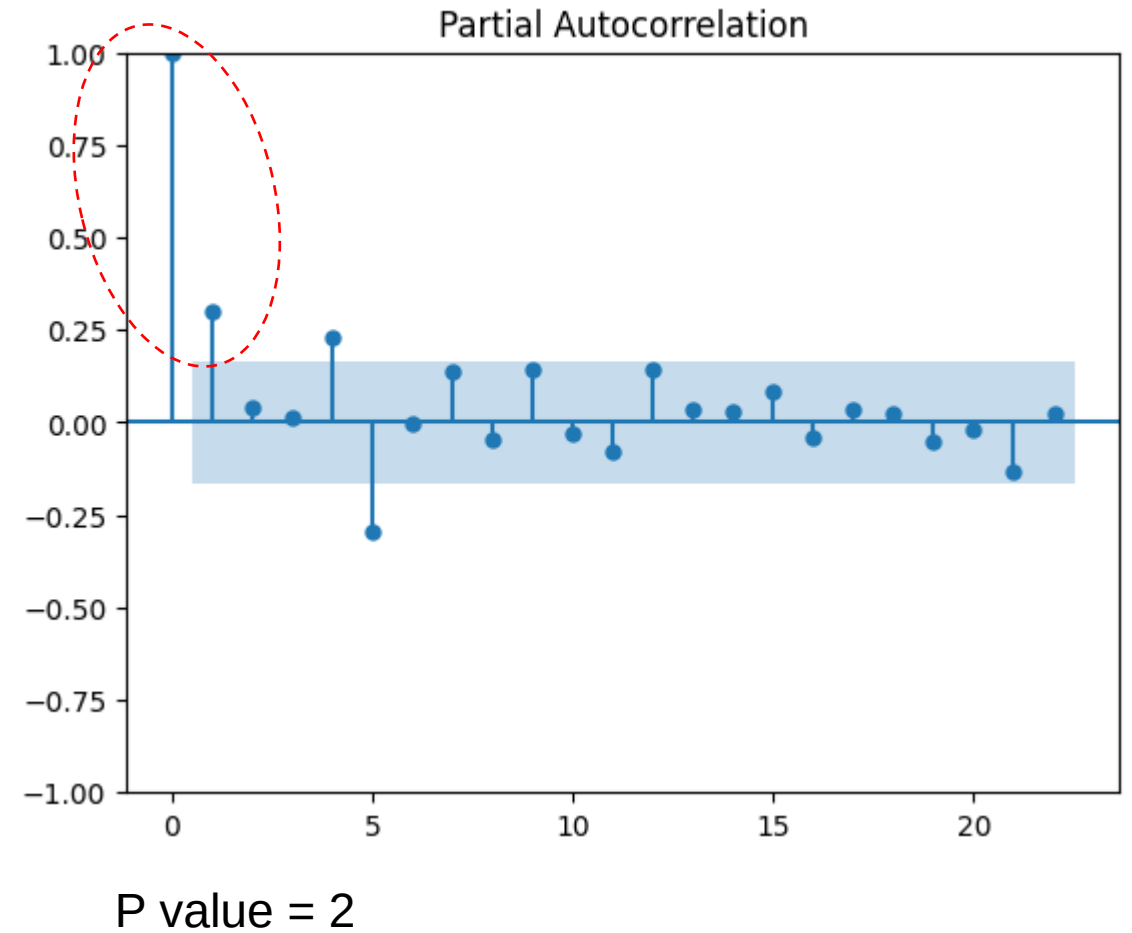
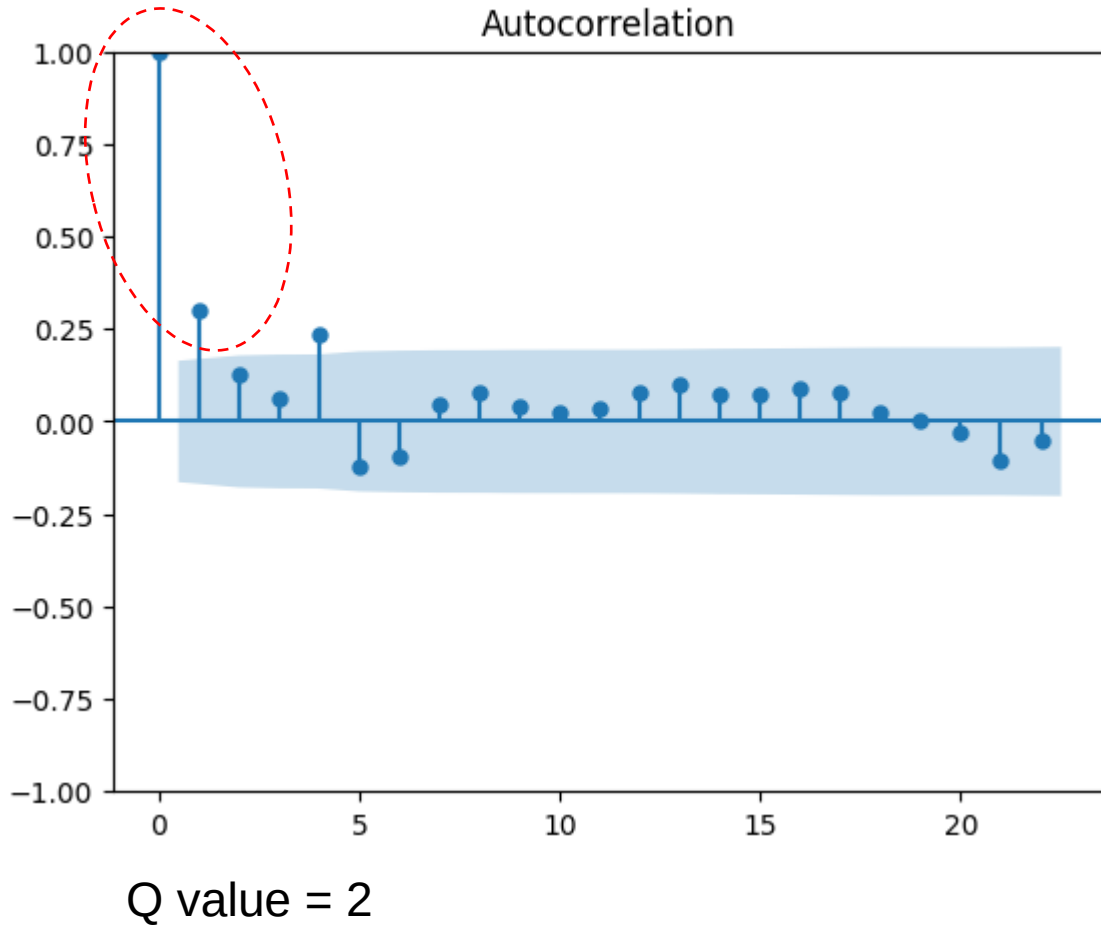


# Assumed seasonality window = 52 weeks

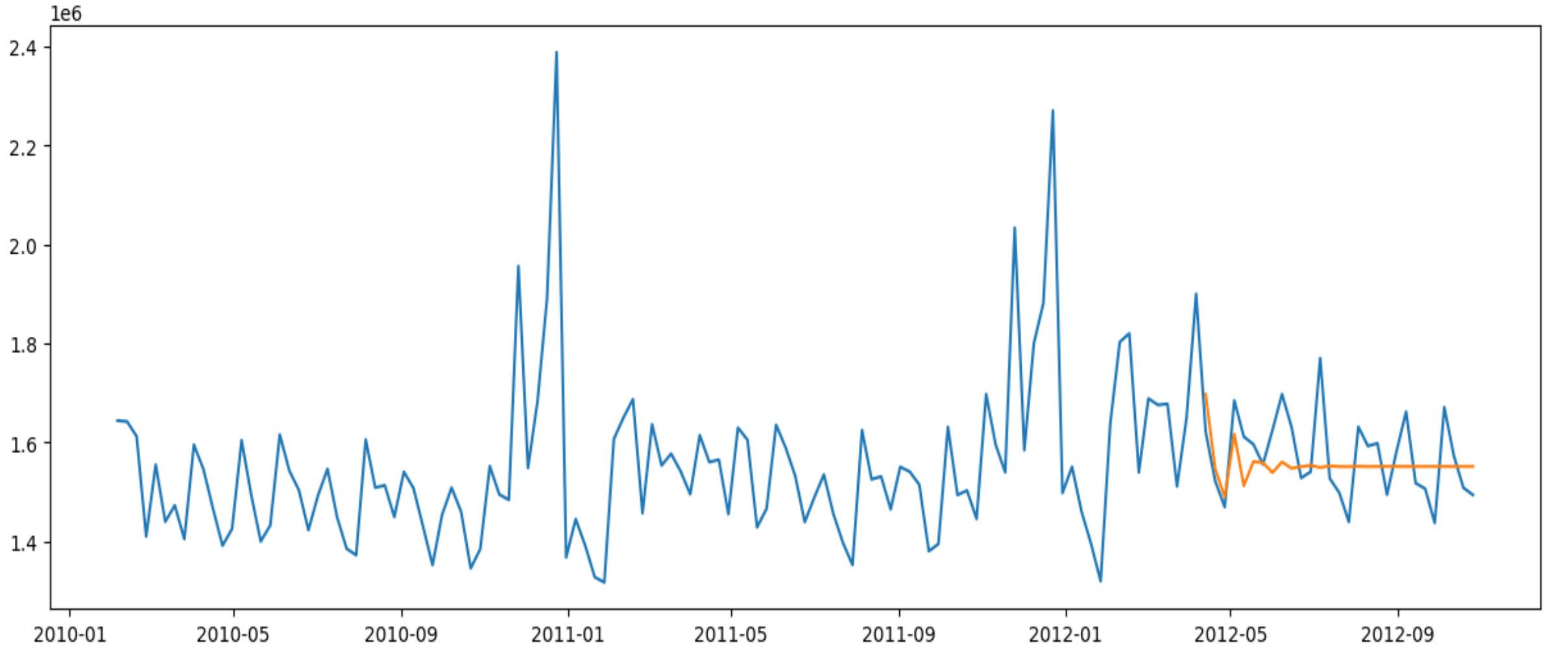
Since plot of mean data gives a straight line with window = 52 weeks



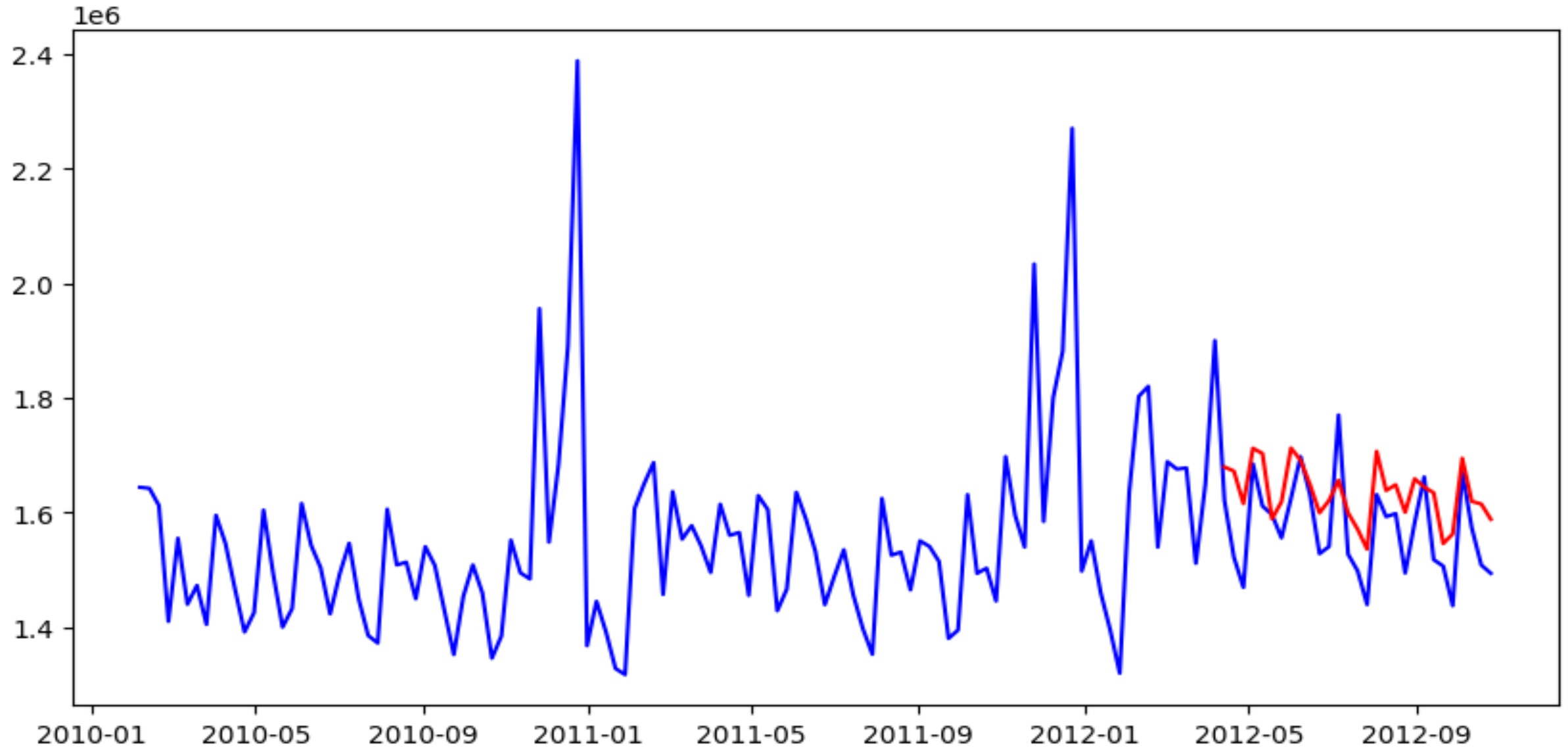
# Conducted ACF & PACF Tests



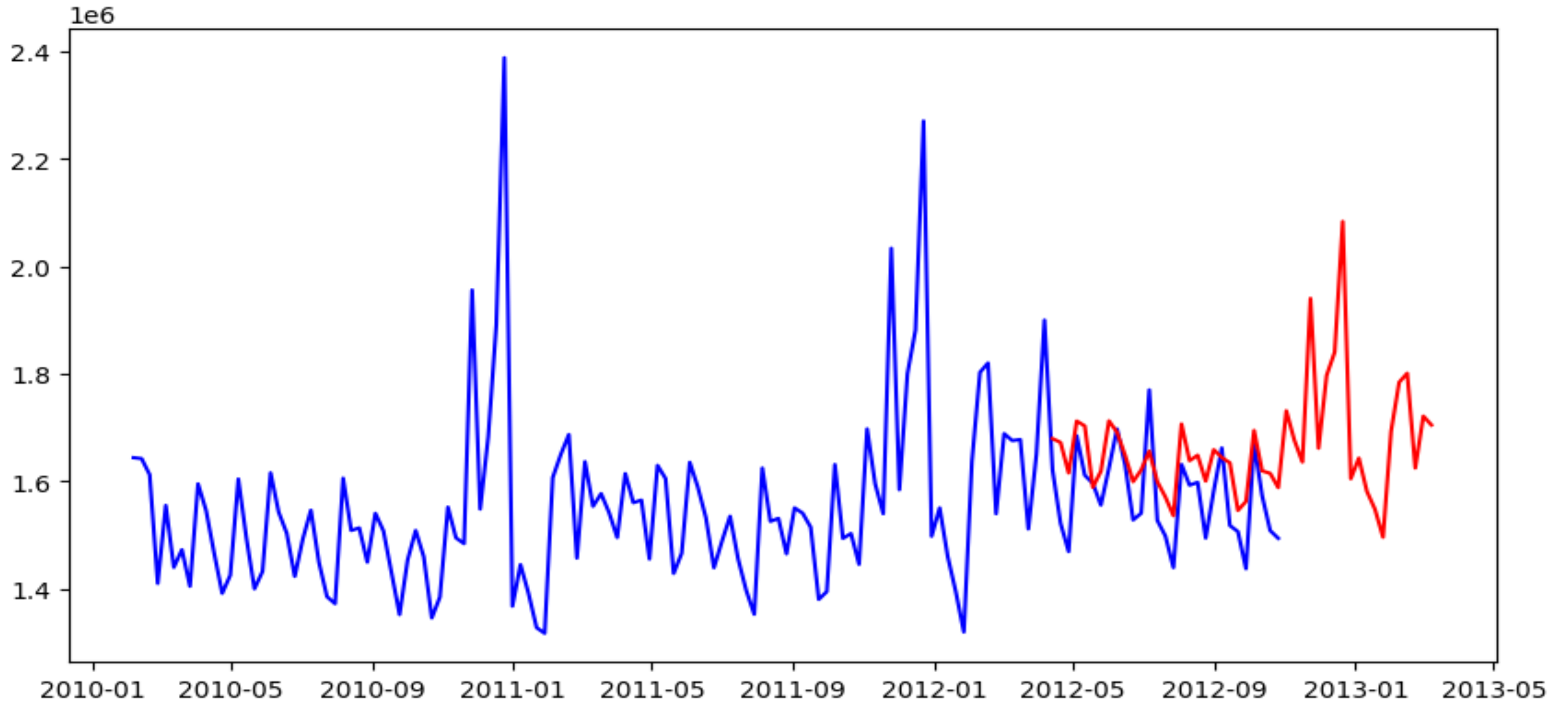
# ARIMA Output



# SARIMAX Output



# SARIMAX Predictions





# Model Evaluation

- For model evaluation technique used was finding RMSE values of both the model and whose RMSE value was less is considered the best model .

## Inferences

- Seasonality-
  - Model has detected and removed seasonality using Dicky Fuller test.
- Discrepancies
  - Certain small discrepancies may happen because
    - Certain months have 5 weeks whereas most months have 4 weeks in the dataset.
    - Although outliers were removed- but it may still impact data prediction.
- Overall
  - SARIMAX seems a good choice for time series prediction.

# Future Potential

- The model and its techniques can be extrapolated for other situations, industries, clients, problems for predicting time based data.

## Conclusion

- SARIMAX is a powerful tool/ ML algorithm to use for predictions
- Data Cleaning is extremely important
- Making the data stationary is very important
- Great insights were drawn from the dataset

# Thanks